WORKSHEET · PYTHON (GR 6–8) BEGINNER: PYTHON GAME CODERS

# Lesson 5: Loops: Repeat Like a Robot

Module 2 | 60 minutes | for, range, while | Grades 6–10 · ages 11–15

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

## 1. Key words

Match each word to what it means. Write the letter.

a. loop ___ A. Repeats as long as a condition is True.
b. for loop ___ B. Makes a list of numbers: range(1, 6) is 1 to 5.
c. range() ___ C. Repeats once for each item, such as each number in a range.
d. while loop ___ D. Code that repeats.

## 2. Plan your program

Before you type it: Times-table printer. Ask which table the user wants with int(input(...)). Then use a for loop with range to print 10 lines like `7 × 1 = 7` up to `7 × 10 = 70`. Write the steps in plain English (pseudocode), then the Python you will use.

## 3. Predict the output: range with a step

What does this loop print? Code: for i in range(1, 10, 3): ↵ print(i)

a. A. 1, 4 and 7 on separate lines
b. B. 1, 2 and 3 on separate lines
c. C. 1, 4, 7 and 10 on separate lines
d. D. 3, 6 and 9 on separate lines

### 4. Which loop fits best?

Write each item in the right group: for loop (I know how many times) / while loop (keep going until something happens).

a. Print a cheer 10 times → _____
b. Print the 9 times table up to 9 × 10 → _____
c. Go through the numbers 1 to 100 → _____
d. Keep asking for the password until it is right → _____
e. Keep playing until lives reach 0 → _____
f. Keep asking until the user types quit → _____

### 5. Plan your program

Before you type it: Debug it! This countdown never stops. Fix it so it prints 5, 4, 3, 2, 1 and then `Go!`, exactly. Write the steps in plain English (pseudocode), then the Python you will use.

**How did it go?** Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can repeat code with a for loop and range().
☐ I can count up, count down and count in steps.
☐ I can use a while loop and make sure it stops.
☐ I can fix an infinite loop.